Avni Sanghvi

DSC 450 | Database Processing for Large-Scale Analytics| Spring 2026 | Assignment 4

Part 1

The solution of all sub-parts of Part 1-A is in the .sql file submitted.

The output screenshots for Part 1-A's 6,7 and 8 are below:

1-A-6:

Query Result

SQL | All Rows Fetched: 12 in 0.07 seconds

	ANIMAL_NAME	ZOO_KEEPER_ID
1	Galapagos Penguin	1
2	Emperor Penguin	1
3	Alpaca	1
4	Sri Lankan sloth bear	2
5	Grizzly bear	2
6	Giant Panda bear	2
7	Siberian tiger	3
8	Bengal tiger	3
9	South China tiger	3
10	Alpaca	3
11	Florida black bear	(null)
12	Llama	(null)

1-A-7:

Query Result

SQL | All Rows Fetched: 3 in 0.063 seconds

	ZOO_KEEPER	MINFEEDINGTIME
1	Rob Schneider	0.25
2	Jim Carrey	0.25
3	Tina Fey	1.5

1-A-8:

Query Result

SQL | All Rows Fetched: 10 in 0.05 seconds

	ASSIGNMENT_DATE	ZOO_KEEPER_NAME	ANIMAL_NAME
1	01-Jan-2000	Jim Carrey	Galapagos Penguin
2	01-Jan-2000	Jim Carrey	Alpaca
3	01-Jan-2000	Rob Schneider	Siberian tiger
4	02-Jan-2000	Jim Carrey	Emperor Penguin
5	02-Jan-2000	Tina Fey	Sri Lankan sloth bear
6	03-Jan-2000	Tina Fey	Giant Panda bear
7	03-Jan-2000	Rob Schneider	Bengal tiger
8	04-Jan-2000	Tina Fey	Grizzly bear
9	04-Jan-2000	Rob Schneider	Alpaca
10	05-Jan-2000	Rob Schneider	South China tiger

Part 2

A. You are given a following schema in 1NF:
(First, Last, Address, Job, Salary, Assistant) and the following set of functional dependencies:

First, Last -> Address
Job -> Salary, Assistant

Decompose the schema to make sure it is in Third Normal Form (3NF).

→ To 2NF:
Table 1: Employee_location(First, Last, Address); 'First' and 'Last' refers to Table3.First and Table3.Last respectively
Table 2: Job_Details(Job, Salary, Assistant); 'Job' refers to Table3.Job
Table 3: Employee_Job(First, Last, Job)
Note: Foreign keys to the table are written in red font.

These three tables do not have any transitive dependencies and hence it is already in 3NF.

B. Write the necessary SQL DDL statements (CREATE TABLE) to define these the tables you created

→ SQL DDL statement for Table 3:

```
CREATE TABLE Employee_Job
(
  First VARCHAR(15),
  Last VARCHAR(15),
  Job VARCHAR(15),

CONSTRAINT Employee_job_PK
```

```
    PRIMARY KEY(First,Last,Job)
);
```

→ SQL DDL Statements for Table 2:

```
CREATE TABLE Job_Details
(
 Job VARCHAR(15),
 Salary NUMBER(6,2),
 Assistant VARCHAR(30),

 CONSTRAINT Job_Details_PK
   PRIMARY KEY(Job),

 CONSTRAINT Job_Details_FK1
   FOREIGN KEY(Job)
     REFERENCES Employee_Job(Job),
);
```

→ SQL DDL Statements for Table 1:

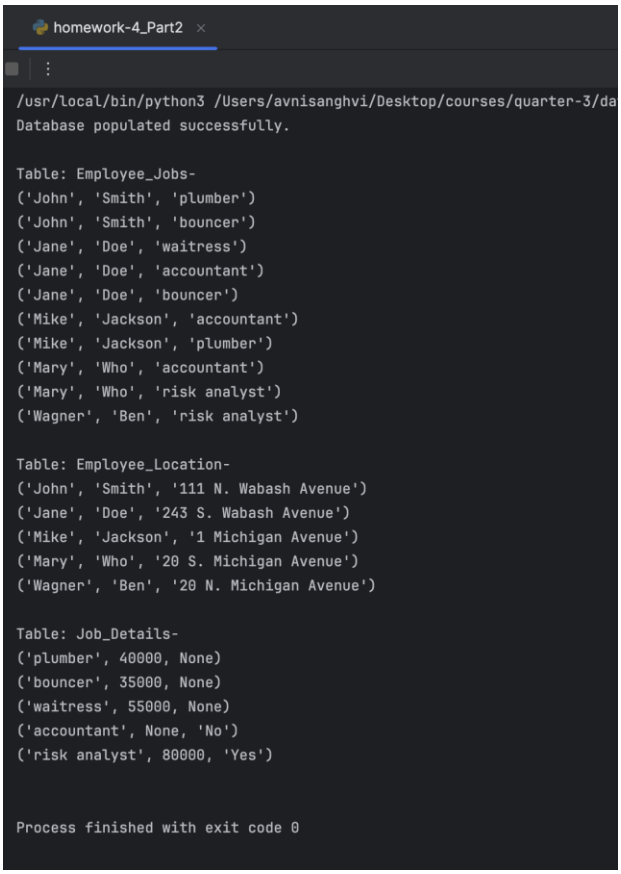
```
CREATE TABLE Employee_Location
(
 First VARCHAR(15),
 Last VARCHAR(15),
  Address VARCHAR(80),

 CONSTRAINT Employee_Location_PK
   PRIMARY KEY(First, Last),

 CONSTRAINT Employee_Location_FK1
   FOREIGN KEY(First)
     REFERENCES Employee_Job(First),

 CONSTRAINT Employee_Location_FK2
   FOREIGN KEY(Last)
     REFERENCES Employee_Job(Last)
);
```

C. Write a python script that is going to create your tables and populate them with data automatically from data_module4_part2.txt (file attached). You do not have to use executemany, your python code can load data row-by-row. Make sure that you are inserting a proper NULL into the database. HINT: You can use INSERT OR IGNORE statement

(instead of a regular INSERT statement) in SQLite to skip over duplicate primary key inserts without throwing an error.

You should include a screenshot of your running code which will include printing the SELECT * FROM [...] each of the tables you have created in Part 2-b and populated in Part 2-c.

For example:
cursor.execute("INSERT OR IGNORE INTO Animal VALUES(?,?,?,?)", [11, 'Llama', None, 3.5]);
would automatically ignore the insert if animal with ID 11 already exists in the database and insert a NULL into the third column. If you use 'NULL' value instead, animal category would be set to the 4-character string 'NULL'



→

→ The python script is attached to the submission folder.

D. Verify that your NULLS are loaded correctly, by finding all jobs with no salary specified using Salary IS NULL condition.

→ The python script is attached to the submission folder

```
Finding all jobs with no salary specified using Salary IS NULL condition.
Rows with NULL as a salary:
('accountant', None, 'No')

Process finished with exit code 0
```